

1 / 5 | Ages 3-4 | 10 minutes

Hand shadow shapes

What we are practising

Observe how an object blocks light to make a shadow.

What you need

Torch / Plain wall

Use daylight shadows on the ground; never look at the sun.

Before you begin

An adult points the torch at the wall in a dim, clear space.

An adult holds the torch; never shine it into eyes. Keep the floor clear.

Try it together

1. Put a hand between the torch and wall and find its shadow.
2. Move the hand closer to the torch while the torch stays still.
3. Return it to the starting place and compare the shadow sizes.

Talk, draw or show your idea

What happened when your hand moved towards the light?

Space to explore

2 / 5 | Ages 3-4 | 15 minutes

Float or sink tray

What we are practising

Make a prediction and compare it with an observation.

What you need

Wide shallow tray / Water / Large wooden spoon / Large metal spoon / Unbreakable empty cup / Towel

Use two different large bath toys if spoons are unavailable.

Before you begin

An adult puts the tray on a towel and adds only enough water for the test objects to float freely without touching the bottom.

An adult stays within reach throughout, prevents drinking, wipes spills and empties the tray immediately afterwards.

Try it together

1. Predict what the wooden spoon will do, then lower it gently.
2. Try the metal spoon and describe where each rests.
3. Try the empty cup upright, then tip water into it and observe any change.

Talk, draw or show your idea

Did everything behave as you expected?

Space to explore

3 / 5 | Ages 3-4 | 10 minutes

Move it with air

What we are practising

Notice that moving air can push light objects.

What you need

Two full sheets of paper

An adult can fan with a thin book.

Before you begin

Fold one sheet into a simple fan; loosely fold the other into a standing tent.

Stay with your child and use large, intact, age-appropriate materials.

Try it together

1. Set the paper tent on a smooth table.
2. Wave the fan towards it without touching it.
3. Try from nearer and farther away and compare the movement.

Talk, draw or show your idea

What pushed the paper even though the fan did not touch it?

Space to explore

4 / 5 | Ages 3-4 | 15 minutes

A gentle ramp test

What we are practising

Compare a change while keeping the toy and release point the same.

What you need

Sturdy cardboard / Two thin books / Large toy car

Use a smooth board with rounded edges instead of cardboard.

Before you begin

Place the ramp on the floor with one end on a book and a clear run-out area.

An adult secures a low ramp; nobody stands on it. Keep the car away from stairs.

Try it together

1. Release the car from the top without pushing and see where it stops.
2. Add the second book under the same end.
3. Release from the same mark and compare; repeat each slope once.

Talk, draw or show your idea

Did the car travel to the same place each time?

Space to explore

5 / 5 | Ages 3-4 | 15 minutes

From ice to water

What we are practising

Observe melting and describe solid and liquid forms of water.

What you need

Large piece of ice / Shallow unbreakable tray / Towel / Paper / Large crayons

Watch condensation droplets on the outside of a cold sealed bottle instead; explain they come from air.

Before you begin

An adult places the ice in the tray on a towel.

Adult supervision throughout; no tasting, prolonged ice contact or hot water. Empty the tray after use.

Try it together

1. Look at the ice and describe its shape without putting it in the mouth.
2. Leave it at room temperature while drawing what you see.
3. Check several times over ten minutes and describe any water that appears.

Talk, draw or show your idea

Where did the little puddle come from?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Hand shadow shapes

A little hint

Compare how much wall the shadow covers.

Answer / what to notice

With the wall and torch fixed, moving the hand towards the torch usually makes its shadow larger.

The hand blocks some light; its position changes the blocked area on the wall.

Make it easier

Watch one adult-made shadow and copy its hand shape.

Take it further

Predict which position makes the smaller shadow, then test.

Another day, away from the screen

Trace an outdoor object's shadow with your finger.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Float or sink tray

A little hint

Look at whether the object stays at the surface or rests below it.

Answer / what to notice

Record what happens: a wooden spoon often floats and a metal spoon usually sinks; cup behaviour depends on its material and trapped air.

Material, shape and trapped air can all affect floating; size alone does not decide it.

Make it easier

Test just the two spoons.

Take it further

Test the cup again and explain what changed.

Another day, away from the screen

Notice floating objects in a picture book.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Move it with air

A little hint

Feel the moving air on your own hand.

Answer / what to notice

Air moved by the fan can push the paper; results depend on distance and how strongly you fan.

Air is hard to see, but we can observe what moving air does.

Make it easier

An adult fans while the child watches and describes.

Take it further

Keep the distance fixed and compare gentle and stronger fanning.

Another day, away from the screen

Watch leaves move in the breeze from a safe spot.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. A gentle ramp test

A little hint

Use a hand beside the stopping point as a temporary marker.

Answer / what to notice

The higher ramp often sends the car farther, but bumps and friction can change the result.

A fair comparison uses the same car, surface and release point.

Make it easier

Use one slope and practise releasing without pushing.

Take it further

Try three releases on each slope and describe the pattern.

Another day, away from the screen

Look for real ramps used by wheels nearby.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. From ice to water

A little hint

Look at the shrinking edges of the ice.

Answer / what to notice

The ice melts into liquid water; a large piece may take longer than the activity to melt fully.

Warm surroundings transfer heat to the ice, helping it melt.

Make it easier

Simply point to ice and liquid water.

Take it further

Draw the ice at the beginning and after ten minutes.

Another day, away from the screen

Ask an adult to freeze water again and check it later.